

Institute of Actuaries of India

Subject SA1– Health and Care

November 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

Solution 1:**i) For the insurer****Advantages**

- Reduces claims administration costs (e.g. no pre-authorisation of expenditure, no processing of invoices for care), and so makes premiums cheaper [0.5]
- Only incidence rates of insured events at various durations after retirement have to be estimated [0.5]
- Do not need an assumption about the claim size distribution [0.5]
- Do not need to estimate the expected rate of inflation in care costs [0.5]
- There is less risk for the insurer and so contingency loadings can be lower, resulting in lower premiums [0.5]
- Lower premiums may lead to increased sales volumes [0.5]
- It may be possible to offer budget policies with modest cash benefits (and modest premiums) that are designed to contribute to care costs rather than aiming to provide the total costs, ... [0.5]
- This would widen the market for this type of policy [0.5]

Disadvantages

- There may be pressure on the insurer to provide advice on spending the cash benefits in the care market [0.5]
- If cash benefits are not sufficient to provide adequate care, the insurer may be blamed, particularly if there is any evidence from sales literature or the sales process that the product may have been mis-sold. The 3% increase in benefit post retirement may be perceived as increase at 3% since inception and may create higher scope of mis-selling. [1]
- The estimate of claim incidence rates at various durations (post retirement) may go wrong [0.5]

For the insured**Advantages**

- All other things being equal, premiums will be less than for care benefits because contingency loadings will be smaller (see under benefits for the insurer) [0.5]
- The insured can choose to spend the money on anything, and so this provides the flexibility of choosing formal care to complement any informal care that may be available. [0.5]
- The 3% p.a. increase may provide a good protection against inflation. However, the protection is post retirement and till retirement it is not protected. [0.5]

Disadvantages

- The amount of cash may not be sufficient to cover the full cost of care unless the sums insured are kept up to date [0.5]
- The insured or relatives will be responsible for buying care when the insured is frail and much less able to make effective decisions [0.5]

The insured or relatives will have no experience of buying care, and so may not be aware of all the available alternatives, nor of prices of care etc	[0.5]
The 3% p.a. increase may not provide a good protection against inflation as health care inflation is generally exceeds general inflation.	[0.5]
The 3% increase in benefit post retirement may be perceived initially as increase at 3% since inception and may create higher scope of mis-selling.	[0.5]
There should be regular top-ups till retirement according to the revised estimation as the protection of 3% is available post retirement only.	[0.5]

[Max 8]**ii) Pre-funded policy**

It is a protection insurance policy.	[0.5]
The policy is bought when the insured is in good health.	[0.5]
The benefit can be cash or provide indemnity against the costs of care.	[0.5]
Level premiums, or premiums that increase in a pre-determined way, ...	[0.5]
... are paid regularly until death, the expiry of the policy term or until benefits are paid.	[0.5]
Alternatively a single premium may be paid at the outset.	[0.5]
There is a possibility that the insured will pay premiums and receive no benefits.	[0.5]
The most basic products do not offer a surrender benefit.	[0.5]
In some cases where regular premiums have been paid for a minimum period a paid-up benefit may be available, which would normally reduce the benefit amount.	[0.5]
The policy may be unit-linked or conventional.	[0.5]
Benefits triggered by disability are tax free.	[0.5]

Immediate needs policy

It can be expressed as a form of impaired life annuity.	[0.5]
The policy is bought when the insured is in poor health, and in need of care.	
Benefits are usually in the form of regular cash income, but which might be designed to meet the expected costs of care.	[0.5]
A single premium is paid at the outset.	[0.5]
The insured will always receive some benefits.	[0.5]
Policies will not acquire a surrender value at any time.	[0.5]
However, some policies may offer a capital guarantee (eg purchase price less benefits received is paid on death). This will increase the purchase price for a given benefit level.	[0.5]
If the policy qualifies as an "Immediate Needs Annuity", payments that are made to a registered care provider for the care of the person covered are generally not taxable in the hands of the insured person.	[0.5]
Otherwise, the insured will be taxed on the income content of the benefit payments.	[0.5]

Advantages and disadvantages for the insured**Pre-funded policy**

+ provides protection against the cost of long-term care for as long as this is required in old age

- + reduces the financial burden on the insured's family and/or protects the insured's assets (eg the family home) which might otherwise need to be used or sold to pay for care
- + there can be a limited premium payment term (eg premiums cease at age 85), which helps the insured to meet the costs more easily
- + premiums may be relatively modest if the policy is taken out at a young age
- however, generally, cover may be expensive, and hence unaffordable, due to the margins required, lack of pre-funded LTCI providers in the market, etc
- the payment of regular premiums will be a long-term commitment for the insured,
- ... which may be at a time when there may be other, more important, demands on the insured's finances (e.g. personal pension, other insurances or mortgage repayments)
- premiums are rarely guaranteed, and so premiums may be reviewed (upwards) every five years, say, which makes it difficult for the insured to budget
- policy will usually have no surrender value and the insured will lose out if he or she needs to lapse the policy – for example, as the result of changing circumstances such as the loss of job

Immediate needs policy

- + helps protect against risk that savings will not be sufficient to provide continuing care for remaining lifetime, or that one's estate will be used up in meeting the cost of care [0.5]
- + specialist products (eg indemnity-based or with different levels of cover) may also protect against increasing cost of care due to further deterioration of health and/or future inflation of care costs [0.5]
- + the policy fulfils an immediate need, and so there is no possibility of premiums being "wasted" because benefits will always be paid [0.5]
- benefits are usually in the form of cash, and so are not guaranteed to cover the total cost of care [0.5]
- the insured will need to find a large sum to pay the single premium... [0.5]
- ... this may require him or her to realise assets or to borrow against the security of assets such as his or her home [0.5]
- the cost of the policy will depend on the financial circumstances at the time it is bought (e. g. premiums will be higher if interest rates are low at the time) [0.5]
- the insured has no control over the time that the policy will be needed, and so may turn out to be short of assets when they are required [0.5]

[Max 16]

- iii) Given that there are substantial volumes of business the company needs to be aware of how the business is performing. [0.5]
- This is so that it can spot possible trends in order to take early action. [0.5]
- The key reason for monitoring is to identify divergences in actual experience from that which was assumed in the contract pricing, as part of the actuarial control cycle for the business. [1]

The product is dependent on Reinsurance terms as 20% of the business is expected to be reinsured. The reinsurance terms are dependent on the experience of the reinsured portfolio. Any change therein as a result of their experience will impact the insurer. Hence, the experience of reinsured portfolio will be monitored with the same seriousness.	[1.5]
The company is particularly interested in whether the contract is proving as profitable as expected ...	[0.5]
... and is giving shareholders an acceptable return on their capital.	[0.5]
If not, the company can consider what it needs to change for existing business (to the extent it can) and for new business.	[1]
On its in-force business it might be able to improve its claims underwriting if that is found to be a problem.	[0.5]
There is little it can do on pricing if terms are guaranteed.	[0.5]
Either way, the company can revise assumptions in its embedded value calculations, to maintain an accurate view of the worth of the business and to recognize profits (and losses) appropriately.	[1]
If the existing contracts are reviewable, then the results can be used to change either the premiums or the benefits given.	[0.5]
However, the scope to do this will to an extent be limited by competitive forces and the need to treat customers fairly.	[0.5]
For new business it will want to use the results of monitoring to help it re-price, where this is shown to be necessary.	[0.5]
The information gained can be used to revise assumptions and produce new premiums.	[0.5]
By monitoring business volumes and mix the company can also gain some information on the marketability and competitiveness of the product by sector.	[0.5]
Monitoring mix is especially important if (as is likely) there is cross-subsidy between policies.	[0.5]
The company also needs to monitor aspects of experience such as duration of claim occurrence post retirement, mortality, expenses and persistency in order to check the appropriateness of supervisory valuation assumptions and for EV reporting.	[1]

[Max 10]

iv) Suitable assets

The company should invest so as to maximise return, subject to acceptable risk.	[0.5]
The benefit payments from the LTCI policies may consist of 3% p.a. increasing benefits that broadly increase in line with general inflation.	[1]
We therefore require further information about the mix of business and how benefits increase, that is durations to retirement for the pre-funded portfolio and duration post retirement for the immediate need business.	[1]
... both during the policy term and during claim payment.	[0.5]
Expenses can also be considered to increase in line with an index, again perhaps salary inflation.	[1]
Premiums will normally be level, but again may increase over time in accordance with the contract wording. These can be netted off the increasing benefit where applicable.	[0.5]
Again, more information on premiums will be needed from the company.	[0.5]

Assets should be in the same currency as that of the liabilities. We need to know whether any of the policies cover overseas policyholders as well.	[0.5]
If there are liabilities in other countries, these could be matched with overseas investments.	[0.5]
For the benefits guaranteed in money terms, a mixture of suitable investments with matching terms would a suitable match.	[0.5]
For example, a variety of fixed-interest bonds and cash.	[0.5]
For the increasing benefits, assets that increase similarly, such as index-linked gilts, would be suitable.	[0.5]
An alternative would be other assets expected to give a real return, such as equities or property, ...	[0.5]
... although the volatility and marketability of these would need to be taken into account.	[0.5]
We should therefore like to know the size of the insurer, and its level of investment expertise to see whether these alternatives would suit.	[0.5]
The company is expected to have good expertise and would probably want a significant proportion of volatile assets.	[0.5]
The size of the insurer's free surplus will also have a big influence on the desirability of more risky assets, ...	[0.5]
.. since a well-financed company could afford a more mismatched and risky investment approach. However, investments in risky assets should be more for the pre-funded liabilities than the immediate need liabilities.	[0.5]
As the company is a proprietary one, it has a duty to its shareholders to sustain and enhance the return it achieves on its assets backing the free surplus. This would imply some investment in equities and property.	[1]
We should also like to know what legislative restrictions there are on the investments we can make.	[0.5]
As the company is growing rapidly, provided it remains solvent, it can use incoming premiums to pay claims.	[0.5]
However, future premium income is not guaranteed and so caution should be exercised.	[0.5]
We should like to know the recent claims experience of the insurer. If the outgo is highly variable, a more cautious approach should be taken, retaining a high proportion of liquid assets.	[1]
Also, the extent of guarantees should be known, as a large proportion of guaranteed business may make net outgo more volatile.	[0.5]
Short-dated gilts and cash could be held to back the capital requirements. This would reduce the exposure to market risk and therefore limit the size of the Solvency Capital Requirement (SCR).	[1]
The need to minimise capital requirements is less important to a company with a large free surplus, so we should like to know the size of the SCR in comparison to the size of the free surplus.	[0.5]
We also need to know the tax position of the company, since this will also influence the asset mix.	[0.5]
The results of any ALM exercise would be useful.	[0.5]

[Max 14]

v)

The fact that the expense and morbidity items are negative is worrying.	[0.5]
---	-------

This is showing that claims are heavier than assumed and/or that claim amounts are more than assumed (eg because more claimants are there after higher durations post retirement.	[1]
Likewise, it is showing that expenses are higher than assumed, so provides some evidence that expenses are high...	[0.5]
... Especially if the company is adopting an active approach to the EV and so the basis does reflect current expense experience.	[0.5]
The negative operating variances imply the current year experience is adverse and no indication as to whether the future best estimate assumptions are changed or not.	[0.5]
Change premium rates immediately? Premiums rates are derived using future assumptions and the similarly future assumptions are used to calculate the EV. We need to know whether both are same or different along with extent of difference as of now. Accordingly EV assumption changes will impact the premium.	[1.5]
For morbidity experience, the key thing is to get a best estimate of future experience. This would come from:	[0.5]
• A detailed claims experience investigation, split by policy class and duration • Allowance for future trends based on the company's experience over a number of years, CMI data, reinsurer's data, other published info etc.	[1]
Higher claims might have been contributed by the higher proportion of immediate need policies with possibly higher durations since retirement who are claiming already.	[0.5]
The variance was positive last year and become negative this year only. It does not look adverse at the first hand unless proved otherwise.	[1]
For expenses, we need a detailed expense investigation, splitting initial, renewal and claim expenses by class.	[1]
These expenses should be compared with the expense loadings in the premium rates.	[0.5]
Even if morbidity or expenses do differ from assumed, we may not automatically change the rates:	
• consider the materiality of the difference and hence the amount by which premiums would change • we need to look at more than just one year in isolation and consider the long-term trend • what policy classes are we talking about – the policies may have been sold many years ago, so there is nothing we can do about it • Consider competition and impact that the change would have on sales volumes.	[1]
Only 2 years have been passed so the higher new business along with the high initial set up cost might have been contributed to the higher expenses.	[0.5]
It seems that it is too early to initiate premium rate change discussions or changing assumptions. Particular care should be ensured before taking it up with the Government Authorities.	[1]

[Max 12]

Solution 2:

- i) Mergers and acquisitions aim at facilitating two companies achieve certain financial objectives. Some of them are listed below:

I. Growth:

The most common reason for merger is growth.

II. Synergy:

Another reason for merger is synergic benefits. This is the most commonly used word in Mergers and Acquisitions, for increasing performance and reducing cost of operations by combining the business activities. Two businesses merge together if they have complementary strengths and weaknesses i.e. it follows the financial maths $1+1=3$. This shows that the value of the two firms combined is much more than the two of them operating independently.

There are two forms of synergies derived from:

Cost economies: They help eliminating duplicate cost factors such as redundant personnel and overhead. These lead to lower per unit costs.

Revenue enhancement: This is when one company's marketing skills combine with the other company's research process to significantly increase the combined revenue.

Synergies are positively correlated to Mergers and Acquisition. This means higher the synergy, higher the target gains as well as the acquiring firm's shareholders benefits

Further, there are three types of synergies:

Operational synergy:

This is achieved by earning operational profits which is done by linking assets of the companies together to be used for various purposes. Operational synergy can be achieved by one company by opting for a merger or an acquisition by eliminating its weakness i.e. for example if a company has a strong distribution, it can acquire a company with a good product mix and technical know-how thus resulting in the company to be stronger. Operational synergy can be achieved by economies of scale or economies of scope.

Financial synergy:

This includes when two companies after a merger or an acquisition achieve high return on equity, right to use larger and cheaper capital market. Mergers and acquisitions also provide tax benefits, which is a financial synergy. When the capital of two companies is combined and results in the reduction of cost and a higher cash flow that is also called financial synergy. It is stated that "financial

synergy, on average, tends to be associated with more values than do operational synergies”

Managerial Synergy:

When two companies come together it is possible that one of the company's has better and well skilled managers than the other company. Thus, the managerial synergy helps in forming a new firm with expertise and thus leading to an improved performance of the company.

III. Diversification:

Diversification is when that a company goes through a Merger or Acquisition with a company which is from an unrelated industry.

IV. Economies of scale & Economies of scope :

Size is one of the important factors in M&A. A larger company benefits more from a merger in the form of cost reduction than a small company. The purchasing power and the company's negotiation power improves after the merger i.e. the larger the company higher the chance to negotiate the price of products with suppliers and to ensure to not spoil the relations with the suppliers although the orders maybe inbuilt. This basically concludes that new entity reduces the duplicate operations lowers costs thus higher profits. The economies of scale refers to the average unit cost of production going down as production unit increases. The economies of scale is the goal of the horizontal and conglomerate M&A.

An economy of scope implies higher the number of products the less is the cost of production. The feature of economies of scope is more suitable for vertical M&A in seeking vertical integration. In addition, complementary resources between two firms are also the motive for M&A. It means that smaller firms sometimes have components that larger ones need, so the large company's acquisition of the small company often take place.

V. Increase market share and revenue:

M&A leading to an increased power of the new entity in the market. This helps in increasing market share. It also improves the investment opportunities of the firm; a bigger firm has an easier time raising capital.

VI. Eliminate competition:

Mergers and acquisitions eliminate the competition and increases firms' market share. A drawback is that shareholders need to be paid a huge amount of premium to convince the other company to accept the offer. It is very common that the acquiring company's shareholders sell their shares which leads to reducing the price the company pays for the target company.

VII. Acquiring new technology:

A large company can buy a small company with unique technologies and develop a competitive edge. This is the need of the competitive market.

VIII. Raising of capital:

Better reputation and credit worthiness with the increase in the size of the company helps in raising the capital easily at any time.

IX. Taxes:

The profitable companies generally buy the companies which are loss making, so that it reduces their tax liabilities.

[12]

ii) The “scheme of amalgamation and transfer of General Insurance business-2011” regulation provides a list of information may need to be submitted to the regulator:

1. Proposed structure of the merger
2. Board resolution of the insurers approving the steps to undertake the transaction
3. Confirmation by the shareholders to pursue the transactions
4. Details as to protection of interests of policyholders, creditors and /or shareholders of the transacting insurers
5. Composition of the board of directors and senior management of the merged entity.
6. Compliance with the guidelines of the corporate governance
7. Approval from other regulatory bodies
8. Shareholder agreement with respect to the management and governance arrangement
9. Actuarial report on valuation of the insurance business of the transacting insurers encompassing the assets, liabilities and solvency position.
10. Products that will be sold by the merged entity
11. Assessment of impact on Distribution Channels / intermediaries and branch network
12. Details of the reinsurance arrangements and strategies
13. Implication on key contracts
14. Issues pertaining to Information technology
15. Projected financial statements of the amalgamated entity as on the date of transfer

[6]

iii)

- I. Products –
 - a. The merged entity will need to decide whether to continue to sell the existing bouquet of products or develop new products.
 - b. A detailed analysis of the existing products of both the companies need to be undertaken
 - c. This also gives the insurer an opportunity to evaluate the entire product range of both the companies and design products that meet the demands of both set of policyholders.

- d. Also, if the company does decide to create new set of products for the merged entity then it will be important to consider whether and how to migrate the existing policyholders to the new products
- e. It will be important for the insurer and the regulator to ensure that the existing policyholder's accrued rights and benefits are protected if they are to be migrated to the new products
- f. If a certain product name sold by SAHI is very popular then the merged entity may decide to sell the same product and migrate the policyholder of the GI company to this product

[3]

II. Employees

- a. Given that the GI insurer is also selling health insurance, it will have employees who are in-charge of running the health product line.
- b. Hence the management will need to decide how to integrate the teams. A decision may need to be made whether to merge the employees who are in charge of health business in the GI company with the SAHI and let it run as a company within the company.
- c. Since SAHI is an independent company in its current form, there will be many departments that will overlap with the GI, e. g. HR, Finance, etc. hence, the management may decide to only move certain set of employees of the SAHI to the GI. There may be lay-offs as a result of the merger.
- d. Also the compensation and benefit structures will need to be aligned and the benefit accrued to-date will need to be protected for those employees
- e. Further, if the SAHI is managing claims in-house while the GI is using a TPA then they may want to move the claims management of the

[3]

III. Systems / IT infrastructure

- a. Given SAHI is a leading health insurer, it must have built strong systems and IT infrastructure to manage the business
- b. Again, a comprehensive review of the technology infrastructure available in both the organization will need to be reviewed.
- c. The objective of IT Systems and infrastructure is to provide a seamless experience of policyholders of both the companies after merger. To that intent they need to make sure that there is consistent and seamless flow of data in various systems of two companies.
- d. If there is overlap in the IT system and software being used by different functions across the two companies, the ultimate aim should be to adopt the one which best suit the purpose and retire the other which results in minimum disruption. There can be a situations when both may be running for some time and then the one is led to sunset.
- e. Another very important consideration in system integration is achieving overall objective of minimum data loss. This may mean building more functionalities in the system that we are planning to retain ultimately.
- f. If both companies are using same software the objective should be how to minimize the overall cost by rationalization of licenses.
- g. If one company is using open software and other proprietary the decision should be based on overall Cost-Benefit-Analysis.

[3]

IV. Distribution channels

- a. The various distribution channels used to source business by SAHI will have to be reviewed
- b. The merged entity can benefit depending on the nature of distributors used by the SAHI.
- c. If both the companies are distributing products through same distributor partner then the terms should be compared and made sure that in future the merged entity takes advantage of 'best of both world'.
- d. In case of any conflict in distribution network in any Geography (for example exclusive rights to third party in certain area) the same should be resolved keeping the interests of all the concerned parties in view.
- e. Distribution plays very important role in the whole ecosystem of insurance, therefore, it should be ensured that they feel positive about the merger, else the major part of the value generation may be at stake.
- f. If any distribution channel is to be closed due to any regulatory restrictions after merger the same should be accounted for upfront in the business case for arriving at financial projections of merged entity.

[3]

iv) This could consist of the size of the free assets calculated on a realistic basis plus goodwill.

[0.5]

An adjustment would be made for any capital that must be tied up to keep the company solvent, such as minimum capital requirements (or prudence within the outstanding claims reserve).

Alternatively, a model could be built to project and then discount future dividends.

[0.5]

This model would allow for future new business and for the "lock in" of capital described above.

[0.5]

We have assumed that the first method is used. Markers should give credit for either approach.

The consulting actuary should provide a range of possible alternatives under suitable assumptions and scenarios.

[0.5]

Free assets

This would be calculated as the total market value of the assets less a realistic estimate of the liabilities.

[0.5]

The liabilities would be calculated as the sum of:

(1) Outstanding claims reserve

We should:

- make a realistic estimate of the money needed to meet future claims that have been reported to the target company
- collect as much data as possible to split the claims by claim type
- allow for claims handling expenses and future claim inflation

[0.5]

[0.5]

[0.5]

- calculate figures net of reinsurance [0.5]
- obtain information to be able to judge the strength of the company's
reserving basis and the length of time for claims to run off. [0.5]

(2) Unearned premium reserve

The unexpired portion of premiums that have already been received, allowing
for deferred acquisition costs and reinsurance premiums. [0.5]

We may also require an additional unexpired risk reserve if there is any evidence
that the premiums charged are inadequate.

(3) IBNR

This represents claims in respect of treatments that have been carried out but
have not been reported to the insurer. [0.5]

It should be small, depending on the insurer's claims notification procedures. [0.5]

Goodwill

Estimate this by applying suitable claim and expense forecasts to the estimated
level of premium income in each future year. [0.5]

Allow for expected sales volumes, mix of business, economic influences (and other
relevant factors). [0.5]

This premium income needs to reflect new business and renewals (i.e. persistency
must be allowed for). [0.5]

The present value of the future profits of this expected future business would
represent the goodwill [0.5]

Use a high discount rate to reflect the uncertainty over the level of future sales. [0.5]

Adjustments for capital

An allowance could be made to the purchase price to allow for capital tied up
within the company – e. g. minimum capital requirements. [0.5]

However, for short-tailed business, such as PMI, the level of this capital is not likely
to be significant. [0.5]

[Max 10]

[40 Marks]
